



INSTITUTE OF INNOVATION

Sample Paper Mid-Semester Examination

Mathematics - Calculus & Combinatorics(only P&C)

Total Marks: 50

Duration: 1 Hours 30 mins

Total Questions: 50 MCQs

Instructions:

- Section A: 25 questions $\times$ 1 marks = 25 marks
 - Section B: 15 questions $\times$ 1 marks = 15 marks
 - Section C : 10 questions $\times$ 1 marks = 10 marks
 - Each question has only ONE correct answer
 - No negative marking
 - Use of calculator is allowed
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SECTION A: (1 marks each)

Q1. What is the range of the function $f(x) = |x|$?

- (A) $(-\infty, \infty)$
 - (B) $[0, \infty)$
 - (C) $(-\infty, 0]$
 - (D) $(0, \infty)$
-

Q2. The function $f(x) = x^3 - x$ is:

- (A) Even function
 - (B) Odd function
 - (C) Neither even nor odd
 - (D) Both even and odd
-

Q3. The value of $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$ is:

- (A) 0
 - (B) 3
 - (C) 6
 - (D) 9
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Q4. A function $f(x)$ is continuous at $x = a$ if:

- (A) $f(a)$ exists
 - (B) $\lim_{x \rightarrow a} f(x)$ exists
 - (C) $\lim_{x \rightarrow a} f(x) = f(a)$
 - (D) $f'(a)$ exists
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Q5. The Maclaurin series expansion of $\sin(x)$ starts with:

- (A) $1 + x + x^2/2! + \dots$
 - (B) $x - x^3/3! + x^5/5! - \dots$
 - (C) $1 - x^2/2! + x^4/4! - \dots$
 - (D) $x + x^2/2! + x^3/3! + \dots$
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Q6. If $y = 5x^4 - 3x^2 + 7$, then dy/dx at $x = 1$ is:

- (A) 14
 - (B) 20
 - (C) 9
 - (D) 26
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Q7. For $f(x,y) = x^3y^2 + 2x^2y$, the value of $\partial f/\partial x$ is:

- (A) $3x^2y^2 + 4xy$
- (B) $2x^3y + 2x^2$

- (C) $3x^2y^2 + 2x^2$
(D) $x^3y + 4xy$
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Q8. A function $f(x)$ is increasing on an interval if:

- (A) $f'(x) > 0$
(B) $f'(x) < 0$
(C) $f''(x) > 0$
(D) $f''(x) < 0$
-

Q9. The critical point of $f(x) = x^2 - 8x + 15$ is:

- (A) $x = 2$
(B) $x = 4$
(C) $x = 8$
(D) $x = 15$
-

Q10. $\int (6x^2 - 4x + 3) dx$ equals:

- (A) $2x^3 - 2x^2 + 3x + C$
(B) $12x - 4 + C$
(C) $6x^3 - 4x^2 + 3x + C$
(D) $2x^3 - 4x^2 + 3x + C$
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Q11. The value of $\int_0^2 x dx$ is:

- (A) 1
(B) 2
(C) 3
(D) 4
-

Q12. The number of ways to arrange 4 different books on a shelf is:

- (A) 12
(B) 16
(C) 20
(D) 24

Q13. The value of $P(6,2)$ is:

- (A) 15
 - (B) 30
 - (C) 36
 - (D) 42
-

Q14. The value of $C(7,3)$ is:

- (A) 21
 - (B) 28
 - (C) 35
 - (D) 42
-

Q15. The number of ways to select 3 items from 5 identical items is:

- (A) 1
 - (B) 3
 - (C) 5
 - (D) 10
-

Q16. The number of arrangements of letters in "PEPPER" is:

- (A) 60
 - (B) 90
 - (C) 120
 - (D) 180
-

Q17. According to Rolle's theorem, if $f(a) = f(b)$, then there exists $c \in (a,b)$ such that:

- (A) $f(c) = 0$
 - (B) $f'(c) = 0$
 - (C) $f''(c) = 0$
 - (D) $f(c) = f'(c)$
-

Q18. If $f(x) = 2x^4$, then $f'(x)$ equals:

- (A) $8x^2$
 - (B) $16x^2$
 - (C) $24x^2$
 - (D) $8x^3$
-

Q19. The function $f(x) = x^4$ is concave up when:

- (A) $x > 0$ only
 - (B) $x < 0$ only
 - (C) All real x
 - (D) No real x
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Q20. The value of $\lim_{x \rightarrow 0} (\tan x)/x$ is:

- (A) 0
 - (B) 1
 - (C) ∞
 - (D) Does not exist
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Q21. If $f(x,y) = x^2y^3$, then $\partial^2 f / \partial x \partial y$ equals:

- (A) $2xy^3$
 - (B) $6x^2y$
 - (C) $6xy^2$
 - (D) $2x^2y^2$
-

Q22. How many 4-digit numbers can be formed using digits $\{1,2,3,4,5,6\}$ without repetition?

- (A) 120
 - (B) 180
 - (C) 240
 - (D) 360
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Q23. If $F(x) = \int_a^x f(t)dt$, then $F'(x)$ equals:

- (A) $\int f(x)dx$
 - (B) $f(x)$
 - (C) $f'(x)$
 - (D) 0
-

Q24. The derivative of $\ln(2x)$ is:

- (A) $1/(2x)$
 - (B) $1/x$
 - (C) $2/x$
 - (D) $\ln(2)$
-

Q25. The number of ways to arrange 6 people around a circular table is:

- (A) 24
 - (B) 60
 - (C) 120
 - (D) 720
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SECTION B: (1 marks each)

Q26. The coefficient of x^2 in the Taylor series expansion of e^x about $x = 1$ is:

- (A) $e/2$
 - (B) e
 - (C) $e/2!$
 - (D) $1/2$
-

Q27. If $u = x^2 + y^2 + xy$, then $x(\partial u/\partial x) + y(\partial u/\partial y)$ equals:

- (A) u
 - (B) $2u$
 - (C) $3u$
 - (D) $4u$
-

Q28. The Jacobian $J(u,v)$ for the transformation $u = 2x + y$, $v = x - y$ is:

- (A) -3
 - (B) -2
 - (C) 2
 - (D) 3
-

Q29. For $z = x^2y$ where $x = 2 \pm 0.01$ and $y = 3 \pm 0.02$, the approximate maximum error in z is:

- (A) 0.20
 - (B) 0.32
 - (C) 0.44
 - (D) 0.56
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Q30. The point of inflection for $f(x) = x^3 - 6x^2 + 9x$ is:

- (A) $x = 1$
 - (B) $x = 2$
 - (C) $x = 3$
 - (D) $x = 0$
-

Q31. The absolute maximum of $f(x) = x^3 - 3x$ on $[-2, 2]$ is:

- (A) -2
 - (B) 0
 - (C) 2
 - (D) 4
-

Q32. For Lagrange's MVT applied to $f(x) = x^3$ on $[0, 2]$, the value of c is:

- (A) 1.0
 - (B) 1.155
 - (C) 1.333
 - (D) 1.5
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Q33. $\int (1/(1 + x^2)) dx$ equals:

- (A) $\ln|1 + x^2| + C$
- (B) $\tan^{-1}(x) + C$

- (C) $(1/2)\ln|1 + x^2| + C$
(D) $-1/(1 + x^2) + C$
-

Q34. The value of $\int_0^\pi \sin^2(x) dx$ is:

- (A) $\pi/4$
(B) $\pi/2$
(C) π
(D) 2π
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Q35. If $F(x) = \int_0^x (x^3) t^2 dt$, then $F'(x)$ equals:

- (A) x^2
(B) x^6
(C) $3x^8$
(D) $3x^5$
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Q36. The value of $\int_0^1 \int_0^2 (x + y) dy dx$ is:

- (A) 2
(B) 3
(C) 4
(D) 5
-

Q37. The number of 4-letter words that can be formed from "MATHEMATICS" is:

- (A) 756
(B) 1680
(C) 2454
(D) 3240
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Q38. The number of non-negative integer solutions to $x + y + z = 10$ is:

- (A) 55
(B) 66
(C) 78
(D) 84

Q39. In how many ways can 9 students be divided into 3 equal groups?

- (A) 252
 - (B) 280
 - (C) 420
 - (D) 560
-

Q40. If there are 5 pigeonholes and 13 pigeons, at least one hole must contain:

- (A) 2 pigeons
 - (B) 3 pigeons
 - (C) 4 pigeons
 - (D) 5 pigeons
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SECTION C: (1 marks each)

Q41. The third-degree Taylor polynomial for $f(x) = \ln(1 + x)$ about $x = 0$ is:

- (A) $x - x^2/2 + x^3/3$
 - (B) $x + x^2/2 + x^3/3$
 - (C) $1 + x - x^2/2 + x^3/3$
 - (D) $x - x^2/2 - x^3/3$
-

Q42. If $u = x^3y + y^3z + z^3x$, which satisfies Euler's theorem for homogeneous functions, then the degree is:

- (A) 2
 - (B) 3
 - (C) 4
 - (D) 5
-

Q43. Using Lagrange multipliers, the maximum value of $f(x,y) = 2x + 3y$ subject to $x^2 + y^2 = 13$ is:

- (A) 10
- (B) 13

- (C) $\sqrt{13}$
(D) $13\sqrt{13}$
-

Q44. For Cauchy's MVT with $f(x) = x$ and $g(x) = x^2$ on $[1,2]$, the value of c satisfying the theorem is:

- (A) 1.2
(B) 1.5
(C) 1.33
(D) 1.67
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Q45. If $F(x) = \int_1^{x^2} \ln(t) dt$, then $F'(2)$ equals:

- (A) $\ln(2)$
(B) $2\ln(4)$
(C) $4\ln(4)$
(D) $8\ln(2)$
-

Q46. By reversing the order of integration, $\int_0^1 \int_x^1 e^{y^2} dy dx$ equals:

- (A) $(e - 1)/2$
(B) $e - 1$
(C) $(e^2 - 1)/2$
(D) $e/2$
-

Q47. The number of derangements of 5 objects is:

- (A) 24
(B) 36
(C) 44
(D) 52
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Q48. In how many ways can 6 people be seated around a circular table if 2 particular persons must sit together?

- (A) 48
(B) 96

- (C) 120
 - (D) 240
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Q49. For $f(x) = x^4 - 8x^3 + 18x^2 - 27$, the number of local extrema is:

- (A) 0
 - (B) 1
 - (C) 2
 - (D) 3
-

Q50. The value of $\int_0^{\pi/4} \ln(1 + \tan x) \, dx$ is:

- (A) $(\pi/8)\ln(2)$
- (B) $(\pi/4)\ln(2)$
- (C) $(\pi/2)\ln(2)$
- (D) $\pi \ln(2)$